

Stock Market Analysis Report

Exploratory Data Analysis & Visualization

Prepared By

Ameni Jabeur

This report analyzes the 100 most active stocks using statistical visualization and exploratory analysis.

The objective is to identify market behavior, price distribution trends, volatility patterns, and relationships between stock variables.

Executive Summary

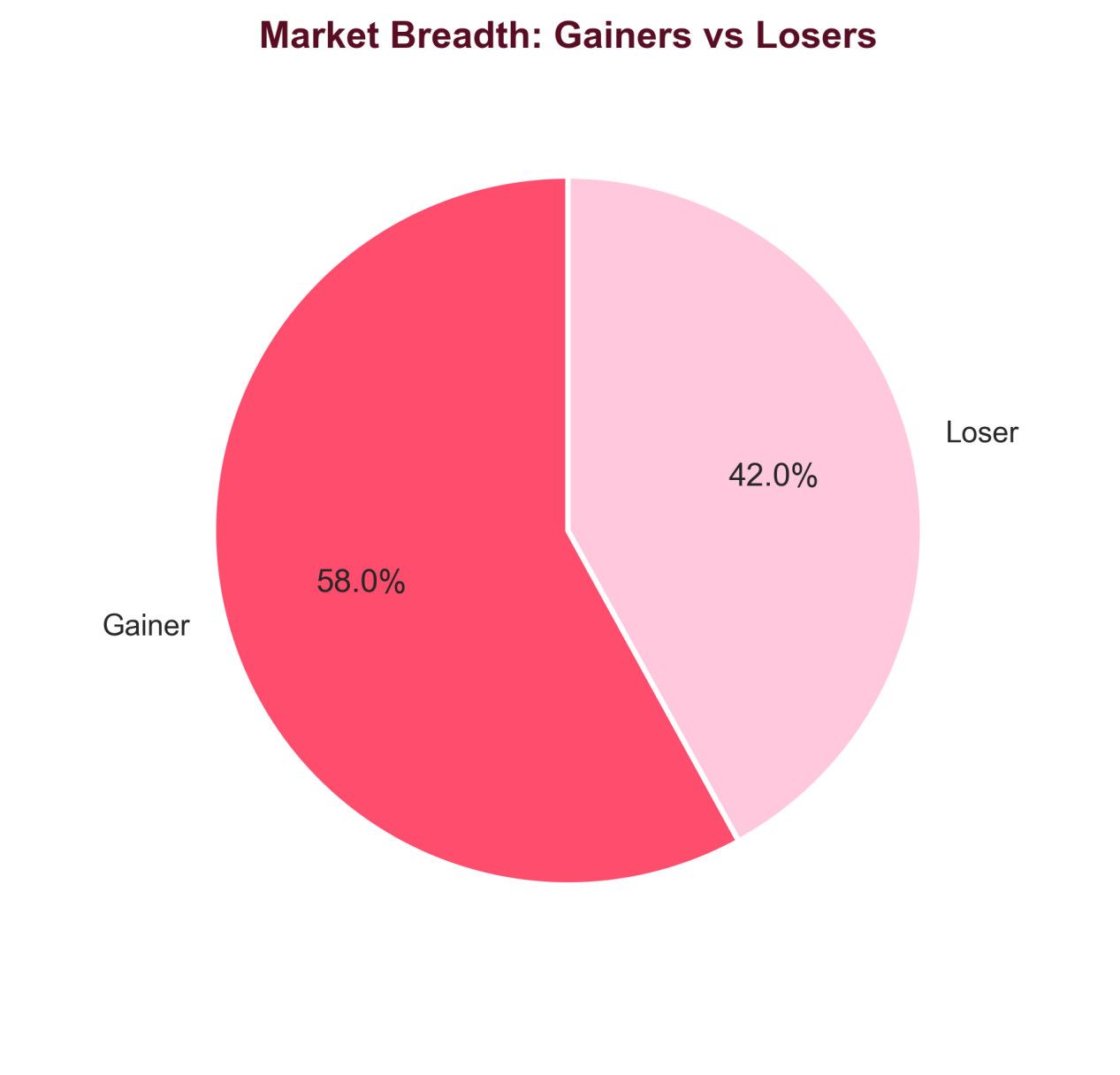
The dataset reveals a market dominated by lower-priced securities, while a small number of expensive stocks significantly influence the overall average stock price. Correlation analysis shows weak relationships between stock price and percentage movement, suggesting that both low-priced and high-priced securities experience volatility independently. The visualizations also demonstrate an active market session containing both gainers and losers with moderate dispersion across percentage changes.

Key Metrics

- Total Stocks Analysed: 100
- Average Stock Price: \$72.73
- Median Stock Price: \$24.30
- Highest Stock Price: \$902.71
- Average Percentage Change: 0.11%
- Number of Gainers: 58
- Number of Losers: 41

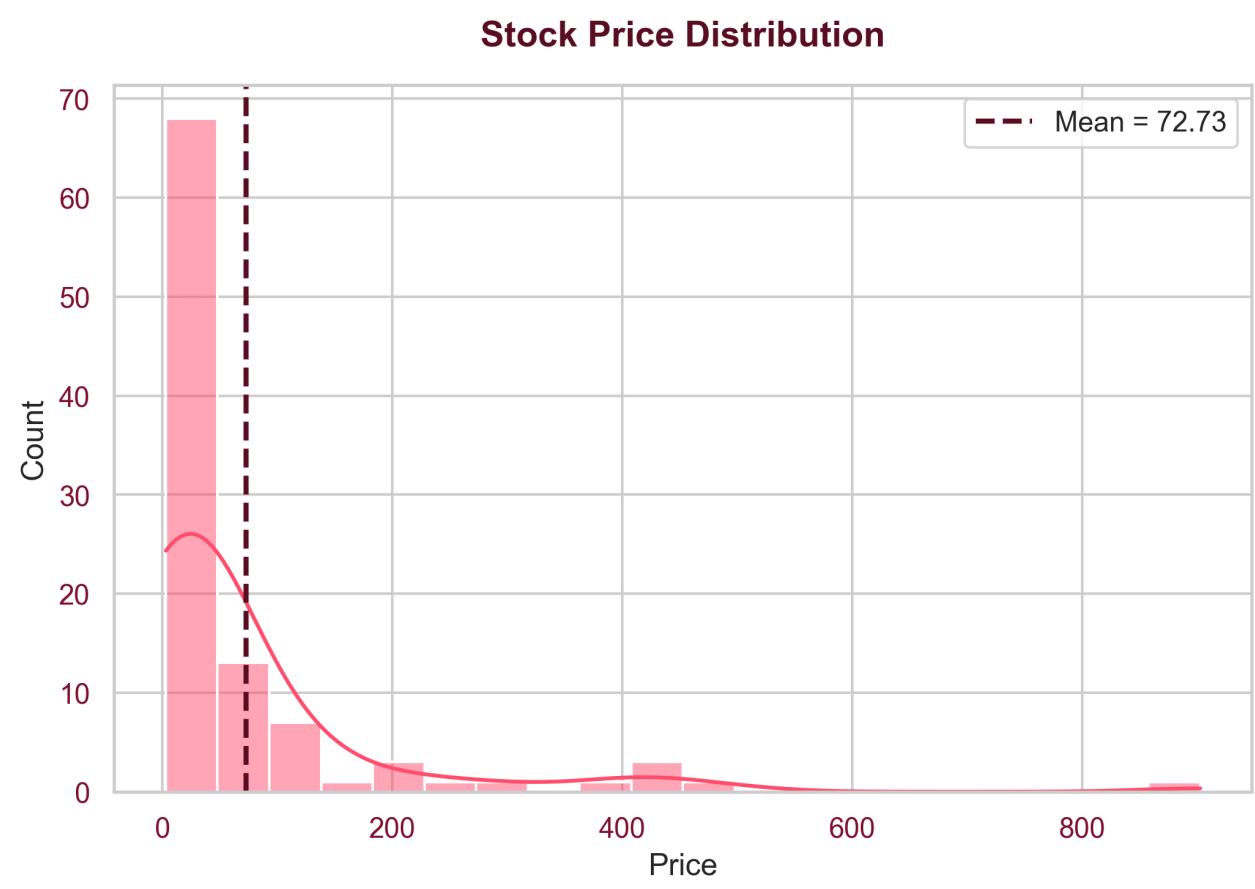
Market Breadth Analysis

The market session contained both gainers and losers, with gainers slightly dominating total activity.



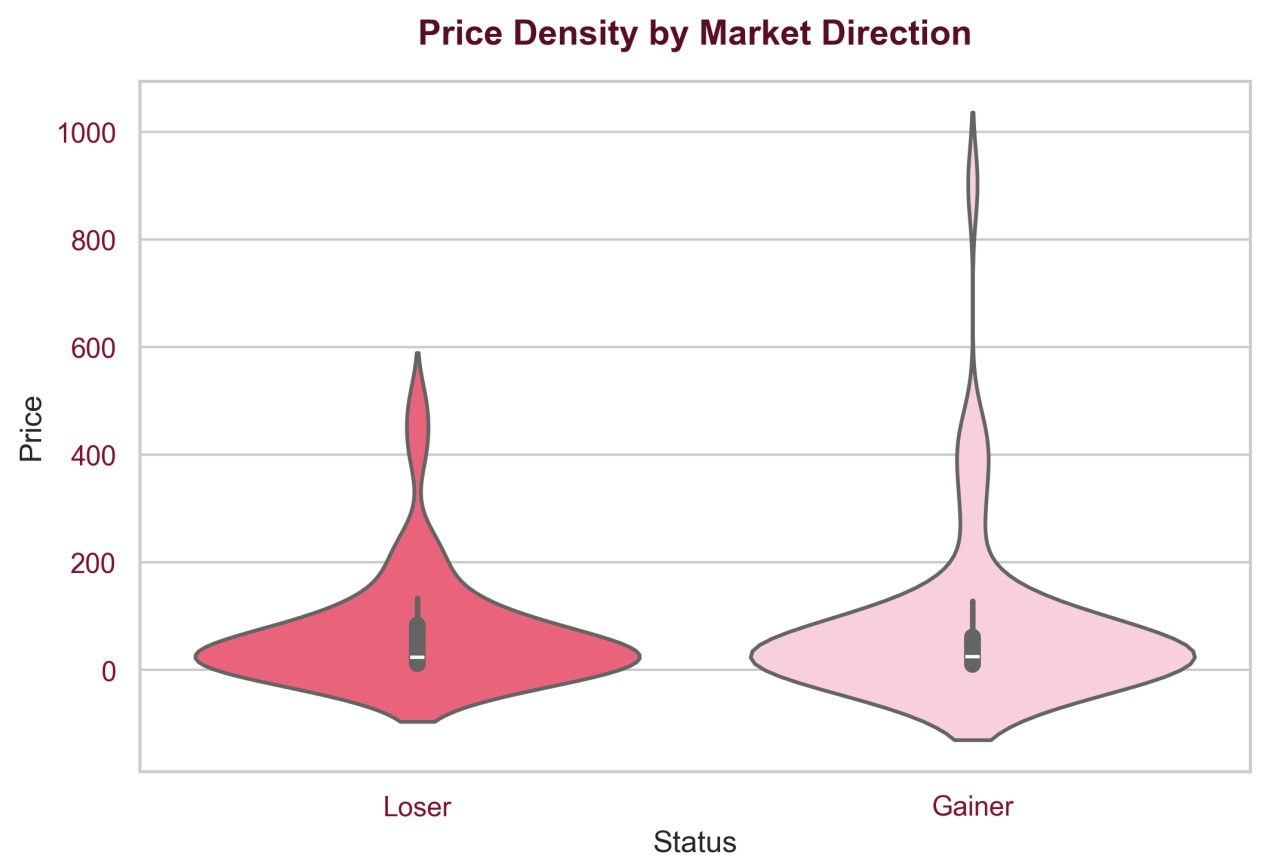
Stock Price Distribution

The distribution is highly right-skewed, meaning a small number of expensive stocks strongly influence the market average.



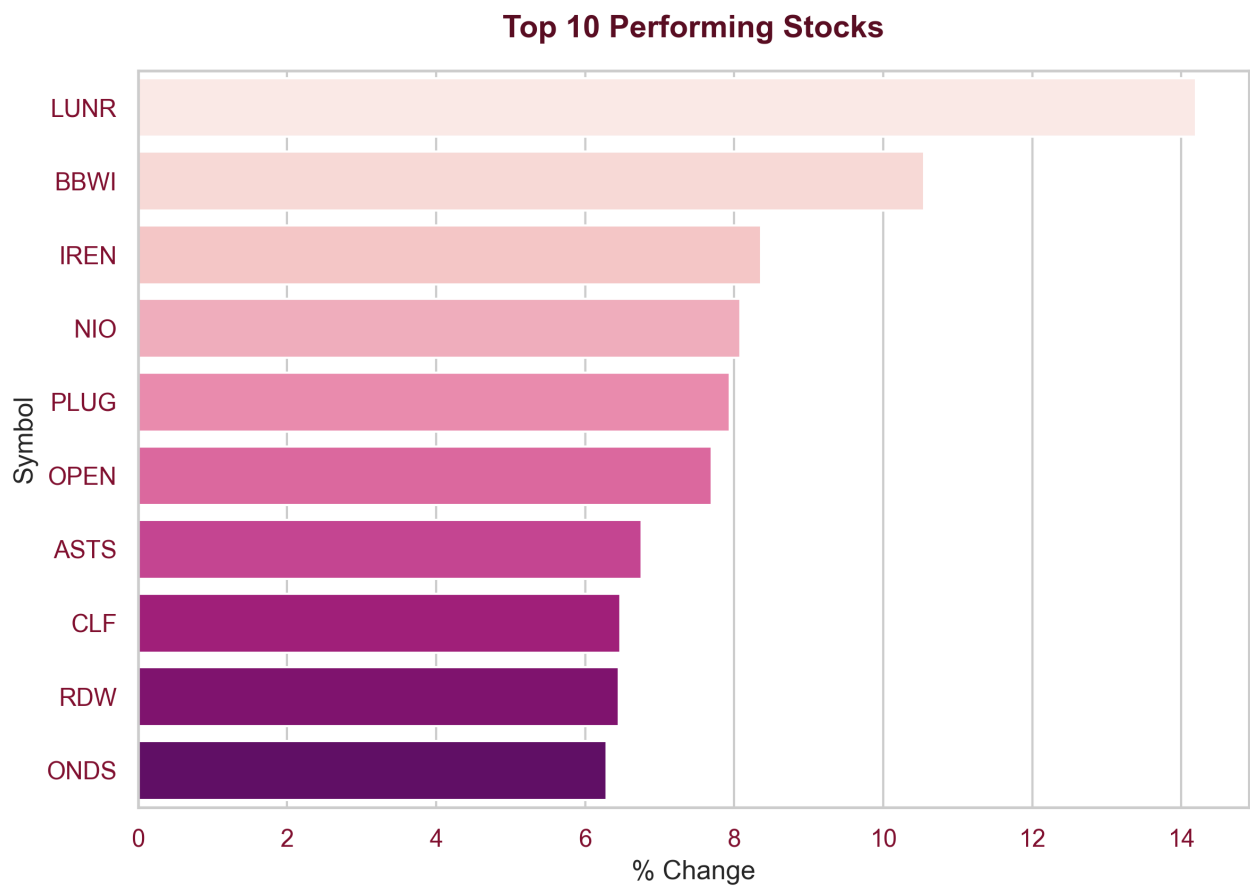
Price Density Comparison

Price distributions overlap significantly between gainers and losers, indicating volatility across all price levels.



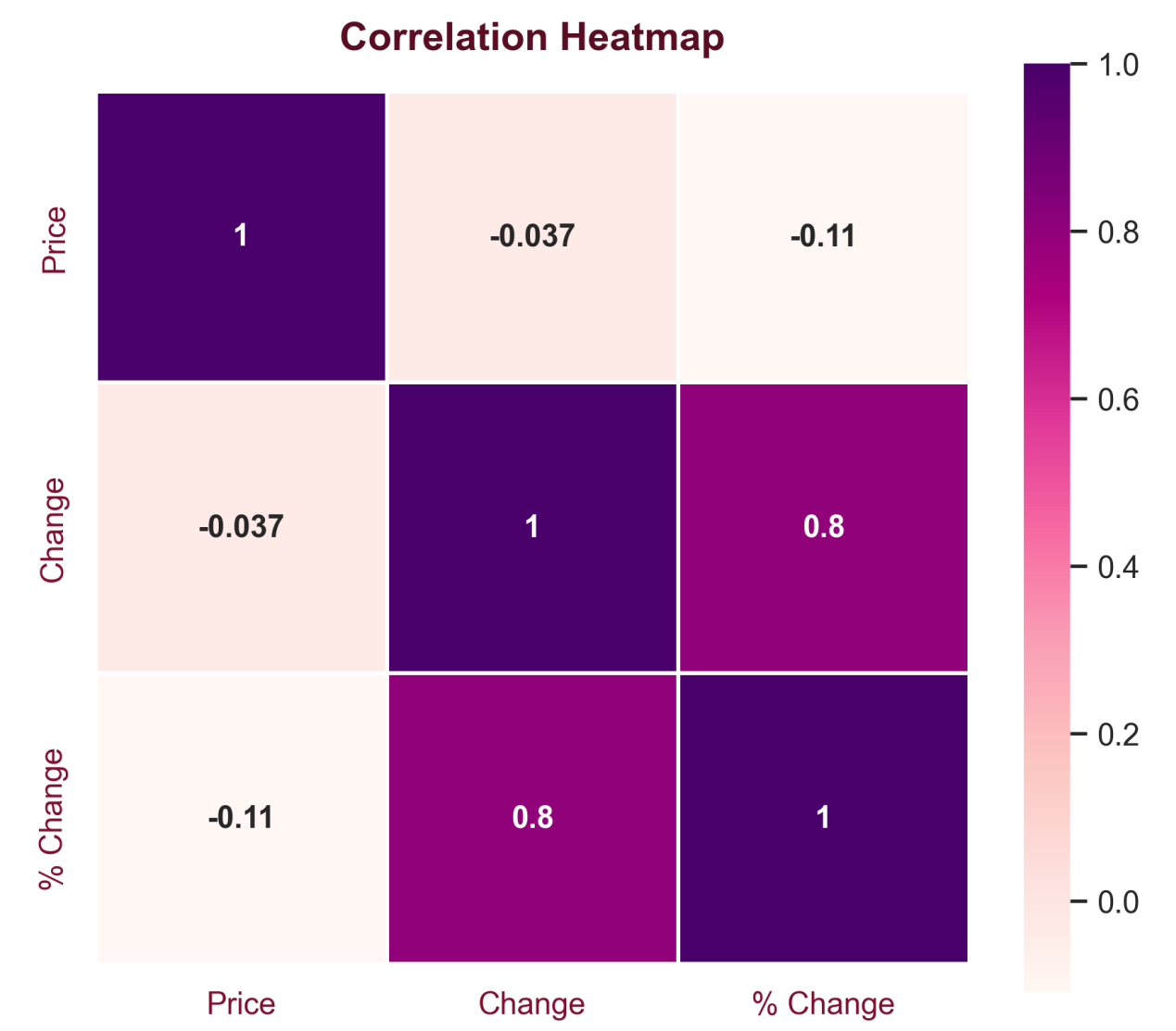
Top Performing Stocks

Several stocks achieved exceptional percentage gains relative to the broader market.



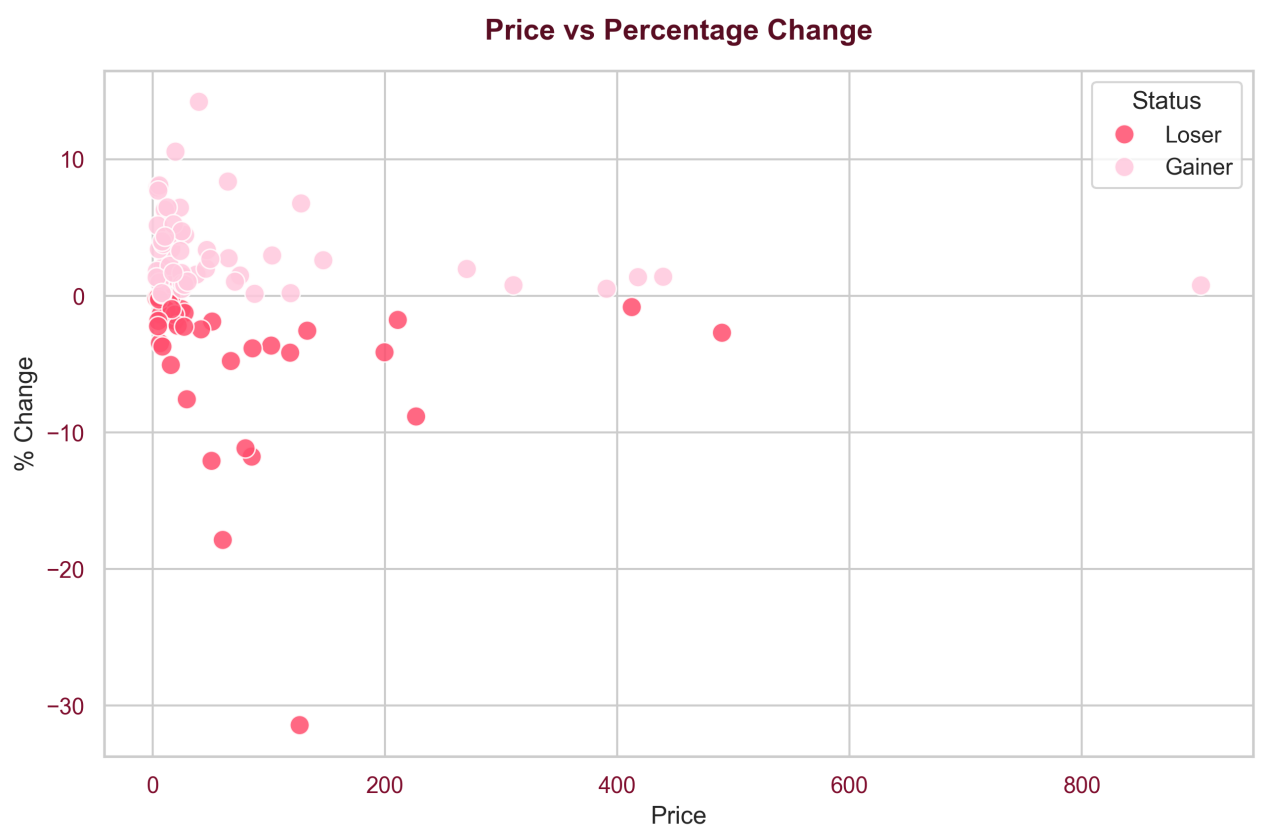
Correlation Heatmap

Correlation analysis reveals weak relationships between stock price and market movement.



Price vs Percentage Change

The scatterplot demonstrates no strong linear relationship between stock price and percentage movement.



Percentage Change Spread

The boxplot highlights market variability and the presence of several statistical outliers.

